Genus-2 goal coverage audit

Snapshot 4bc1aa36a864cbd90f0b8d89a95e5819860459a7

Read-only source and artifact inspection of four original leads and five follow-up lanes. This report distinguishes recorded completed cohorts from remaining source joins, exceptional coverage and new scientific admission. No experiment or proof was rerun and no raw cohort was recomputed. Existing original scope, controls and failures are preserved.

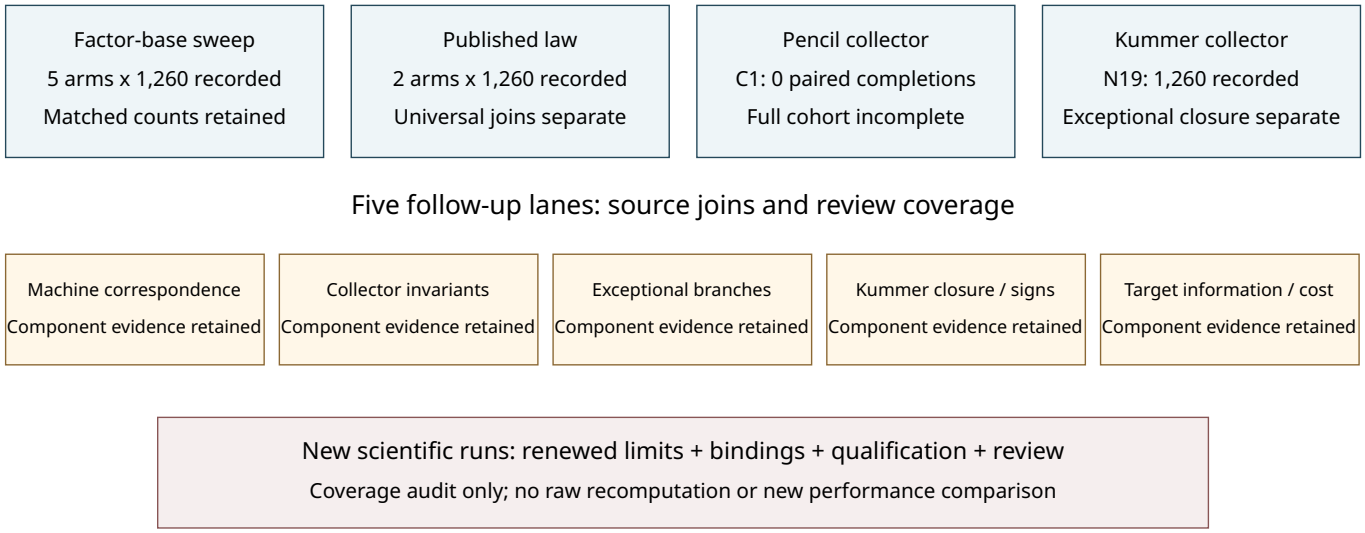


Diagram sources: the nine records in matrix.json; the exact 32 inspected evidence records and SHA256 digests are listed in evidence-bindings.json. This diagram is categorical coverage, not a measurement plot. Cohort numbers are retained-record observations, not new statistical claims.

P denotes research/pencil_lifecycle_20261030/integration/followup-continuation-v2/packets; F denotes the sibling integration/five-followups; R denotes research/projective_law_followup_20261022. Full paths and citations also remain in REPORT.md.

Original leads: factor base and complete law

Factor-base / large-prime parameter sweep

Recorded scope. Five arms fb1-j16, fb2-j16, fb4-j16, fb1-j8, fb1-j32 each have 1,260 source-verified targets. Independent native raw-row repricing reports candidate/rho ratios respectively 2.4488299865553076, 3.0177319451102558, 3.9191927168942735, 2.447173013560974, 2.4532781201923943 at its representative GAE pricing. This requested fixed sweep exists.

Remaining requirement. Preserve the measured parameter set, all price cells and existing replay. Further graph-policy/acceptance experiments are new prospective work; source instrumentation completeness is not supplied by phase conservation alone. Do not relabel this result as absent or launch another identical sweep.

Evidence. `research/four_followups_20261021/results/matched/validator-report.json`

`research/four_followups_review_20261022/VALIDATION.md`

Complete inversion-free genus-2 law

Recorded scope. Wrapper and retained-state arms each 1,260 equal target identities, candidate/rho verified, all five recorded cold phases present. Separate certificate records 5,040 independent source scalar multiplications. Representative recorded ratios 3.7937827176086234 and 4.179297043461173; 36-cell pooled ranges 2.097462349972–21.488710054954 and 1.472855213439–28.015264535504. Retained-state arithmetic avoids routine law inversion but still charges affine boundary normalization: collector is not inversion-free.

Remaining requirement. Universal source-faithful complete dispatcher/class interpretation remains separate from the finished cohort. Later chart/properness/smoothness/infinity-DVR components already exist; reconcile their exact later receipts before redoing geometry. Construct global principal-divisor/class bridge, then source-ordered weighted exceptional composition and invariant closure. Current affine three_leads proofs do not by themselves prove the separate weighted projective_law_gate dispatcher.

Evidence. `R/independent_complete_law/COHORT_RECHECK_20261027.json`

`R/replay-certificate.json`

`integration/four-lead-inventory/law-remaining-obligations.md(locator/context)`

Original leads: pencil and Kummer

Point-driven pencil collector

Recorded scope. Historical standardized archive: 543 distinct workload keys solved, 1 capped-only, 716 absent; 1,506 verified arm-target rows plus 4 cap records are not 1,260 distinct solves. Later p186379 C1 records 7/18 executions, 4/15 measured, 2 pencil timeouts, 0/5 successful comparison pairs, 11 missing.

Remaining requirement. Full 1,260 remains incomplete. Preserve old pencil coverage and C1 separately; no justified additive union count was established here. Remaining enrollment is 1,260 / 22,680 slots (3,780 warmups, 6,300 measured pairs), plus separate gate. Fresh complete costs/ratios require valid admission, qualified runtime/clock, full matched pairs, replay and retained failure charges. Existing p739 timing is invalid for reuse without the specified original-row/equivalence review.

Evidence. `research/four_followups_review_20261022/VALIDATION.md`

`research/pencil_large_prime_20261029/validation/accounting/reports/run-20261030T0543Z-c1-terminal/coverage-counts.json`

`P/continuation-reconciliation/remaining-binding-ledger.json`

Kummer-based walk arithmetic and labels

Recorded scope. N19 completed 1,260 candidate/baseline/rho verifications and 1,260 full-cold charges across 21 models, 36 cells. Cohort receipt terminal exit0. Primary transition validation disabled; fallback_steps=0 and fallback_failures=0. Complete cohort therefore exists but does not exercise exceptional closure.

Remaining requirement. Preserve N19, not rerun just to obtain branch coverage. Complete successful geometric repair/Delta/degree-one closure and source-linked labels beyond fixed controls; implement admissible native recovery/graph semantics and prospective matched whole-cost evaluation of any altered collector.

Evidence. `research/kummer_differential_collector_20261022/n19-labelled-fallback/results/cohort-receipt.json`

`results/analysis/summary.json`

`VALIDATION.md`

Follow-ups: correspondence, invariants and branches

Machine-level field/classifier correspondence

Recorded scope. Independently reviewed bounded word arithmetic, modeled polynomial calls, strict arrays and now actual stored coefficient extraction (5 main +7 controls, 12 producer+3 reviewer readbacks). WellFormed includes residual divisibility/canonical full slots/active leading1; inactive padding may be nonzero. Explicit n=1/no-table, prime and overflow hypotheses.

Remaining requirement. Compose source-ordered complete affine dispatch and successful return storage, then establish applicable input invariant from actual callers. Preserve distinction between native validation, helper p>2, physical Rust/compiler/counter semantics and value model. A separate weighted-law dispatcher still needs its own joins.

Evidence. `P/stored-input-review/status.json+review/REVIEW.md`

`P/native-branch-composition/REPORT.md`

`P/fast-call-review/REPORT.md`

`P/divisor-array-review/REPORT.md`

Collector invariant preservation

Recorded scope. Reviewed field-level mutable find, ranked-forest/global-equation preservation and same-root branch. Literal word same-root endpoint producer has 29 main+11 controls/40 readbacks; word union producer 11 main+4 controls/15 readbacks.

Remaining requirement. Independent source reviews of these word producers remain marked pending. Compose two word finds with distinct-root union, true incoming equation, both anchor orientations and forest termination; add size-vector updates, canonical/sign invariant preservation, overflow/array bounds, returned constraints. Geometric producer must supply a true edge; mathematical graph invariants alone do not establish it.

Evidence. `P/collector-composition/REPORT.md`

`P/state-and-repair-closure/REPORT.md`

`P/storage-forest-adapter/REPORT.md`

`P/native-branch-composition/REPORT.md`

`P/word-find-composition/REPORT.md`

`P/union-and-crt-certificates/REPORT.md`

Exceptional branch classification

Recorded scope. Existing reviewed mixed/same-u/error-exclusion work must be retained. BZero geometry producer records nine named declarations and eight controls, factorization/ordinates/simple cofactor and weighted binding; independent review_complete=false, divisor valuation/class sum false.

Remaining requirement. Independent BZero geometry/dependency review, then finish source-conditioned sequential weighted additions/nonzero weights and full dispatcher coverage. Discharge global divisor-class semantics rather than treating point coordinates or a local exclusion theorem as full addition correctness. Reconcile later chart/stalk proof receipts first.

Evidence. `P/degree22-bzero-geometry-producer/status.json`

`integration/four-lead-inventory/law-remaining-obligations.md`

`laterreviewedsame-u/mixed/wordpacketslocatedinP`

Follow-ups: Kummer and target information

Kummer transition closure / signed recovery

Recorded scope. Non-torsion fixed G-step review: candidate63/63 recoveries vs eager61/63, no repair/Delta coverage. Both graph arms insert35 equations, zero cycles/returns; two added degree-one recoveries are skipped by native degree gate. Separate retained deduplicated forest:175 vertices/61components/114edges/0cycles. Field repair and word row-fold preservation are reviewed; ordinary fiber producer12 declarations remains fidelity-review pending.

Remaining requirement. Independently review ordinary fiber/target orientation, connect native solve4 correctness, chart admissibility, actual runtime matrix-square certificates and projective nonzero/fiber assumptions to geometric recovery. Supply non-two-torsion anchor/prime-order label premises. Degree-one recovery does not add graph edges until a separately justified collector rule exists. Natural successful repair/Delta cost remains unmeasured.

Evidence. `P/kummer-original-gstep-review/status.json`

`P/kummer-graph-review/REVIEW.md`

`P/geometric-forest-review/result.json`

`P/formal-interface-composition/REPORT.md`

`P/native-branch-composition/REPORT.md`

Target-relevant relation information / cost

Recorded scope. Mathematical unique-target criterion: consistent Ax+kb+c=0, b outside range(A), producer8 declarations awaiting review. Offline diagnostic and sign adapter have fixed tests. Raw-sign producer948 matched native rows, actual_collector_streams_checked=0. Reviewed duplicate algebra normalizes BOTH a,b; nonzero delta_b determines target. Retained420 dispositions:218 ineligible,202 compared (174first,28redundant); duplicates give zero deltas in historical analysis. Natural-trace inventory empty.

Remaining requirement. Review formal criterion and native sign producer; implement/bind actual target-local ordered relation-stream capture with exact RHS convention. Preserve first signed labels and verify candidate kG=Q at actual collision. Obtain native pivot/consistency traces, incremental target-rank gain, memory and all insertion/elimination/recovery costs under a prospective protocol. Planting labels validates interface only; cannot supply natural informative collision rate.

Evidence. `F/status.json(locator)`

`P/formal-interface-composition/REPORT.md`

`P/kummer-target-rank-review/REVIEW.md`

`P/target-row-sign-producer/status.json`

`P/retained-duplicate-executed/natural-inventory.json`

`P/collector-composition/REPORT.md`

Admission, costs and independent next tasks

- P/runtime-outcome/status.json: Q0 passed, Q1 program exited0 but strict writable-layer diff failed on /fixtures, /qualification, /qualification/exec-gate. Cleanup complete. Q1 remains failed. - P/runtime-binding-reconciliation/status.json and evidence/binding-ledger.json: timestamped existing image metadata and source-context identities recorded; admitted=false, context_staged=false, new image and qualifier null, ancestor metadata unknown. This does not admit another assembly or scientific run. - P/continuation-reconciliation/remaining-binding-ledger.json retains planning-only 25,200 worker-seconds and absolute deadline 2026-10-31T06:00:00Z, now expired. Spent/remaining/outstanding active-worker accounting null; do not initialize zero. Exact owner closure, actual source/binary/image/protocol/custody bindings, valid new budget/deadline and prospective review remain required. Some older ledger entries (e.g. binary pending) have later source/provenance components and should be reconciled rather than repeated. - New target-information cost fields, natural duplicate yield, replacement collector matched totals and C1 ratios remain null. P/state-and-repair-closure records 17,640 phase-conservation checks over seven frozen arms; conservation of recorded counters is not complete physical instrumentation. - research/pencil_source_work_inventory_20261030/inventory.json names concrete non-affine work: compact reusable buckets; rational-x inventory plus exceptional quadratic roots; sparse rows/path compression. Its bounds are static, measured bottleneck=null. Preserve multiplicity, root/lambda order, reset, pivot order, pending rows, transient caps and complete primitive accounting before comparing.

Useful independent parallel work now (no expired-budget trajectory needed)

1. Independent review of published word-find-composition and word-union producer, followed by one source-ordered distinct-root forest/anchor/size-metadata contract. This closes a different subsystem from affine arithmetic. 2. Independent source-fidelity review of ordinary Kummer fiber and target-identifiability producers in formal-interface-composition; keep mathematical and native implementation acceptance separate. 3. Native solve4/chart certificate specification and input binding for the 105 saved matrix records, using existing word row-fold/nonzero theorem. Reuse saved finite records without claiming general chart coverage. 4. Implement a target-local relation-stream exporter/interface contract and native incremental-rank oracle with exact augmented-RHS sign and byte/operation accounting; first audit saved records, then request prospective natural-stream measurements under new limits. 5. Reconcile the weighted project's later projective/infinity proofs and independent reviews, then construct the missing principal-divisor/class interface; avoid redoing properness/smoothness already built. 6. Source-only compact-bucket or sparse-row candidate with invariant/primitive ledger, independently reviewed before new runtime qualification and cohort protocol. Do not assert measured benefit from static bounds.

This matrix is not a formal evidence-review promotion, independent scalar replay or refreshed cloud-owner check. It changes no original completion criterion and retains prior failures. Root can choose distinct source scopes while execution admission is renewed separately.